

SARANG DEV

Cybersecurity Intern | SOC Analyst | Penetration Tester | AI/LLM Security

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SUMMARY

Computer Science undergraduate targeting a Cybersecurity internship (SOC analyst, penetration testing, vulnerability assessment, AI/LLM security, or security research). Six months of production software engineering experience with React.js and MySQL, paired with hands-on offensive security practice across 15+ CTF competitions on Hack The Box and TryHackMe (top 9% of users). Skilled in web application security, exploit development, cryptanalysis, reverse engineering, network traffic analysis, digital forensics, OSINT, and AI/LLM security (prompt injection, jailbreak testing), with a working knowledge of the OWASP Top 10, OWASP Top 10 for LLM Applications, and secure coding practices.

TECHNICAL SKILLS

- **Security Tools:** Wireshark, Nmap, Burp Suite, Metasploit, Gobuster, ffuf, tshark, Sherlock, exiftool, Cheat Engine.
- **Security Domains:** Penetration testing, vulnerability assessment, web application security, AI/LLM security, prompt injection testing, OWASP Top 10, network security, network traffic analysis, digital forensics, cryptanalysis, reverse engineering, exploit development, privilege escalation, OSINT, threat analysis.
- **Vulnerabilities:** SQL injection, SSRF, IDOR, remote code execution (RCE), WAF bypass, file upload bypass, broken access control, prompt injection, LLM supply-chain attacks.
- **Languages and Databases:** Python, JavaScript, Java, Bash, C++, Solidity, SQL, MySQL, MongoDB, Firebase.
- **Systems and Platforms:** Linux (Kali Linux, Ubuntu), Windows, Microsoft Azure, Docker, Git, GitHub, REST APIs.

SECURITY PRACTICE AND CTF EXPERIENCE

- **AI/LLM Security:** Executed prompt injection and system prompt exfiltration attacks against LLM-powered chatbots and autonomous AI agents, chaining prompt injection with SQL-bypass techniques to extract restricted data; assessed AI model supply-chain risk (malicious pickle deserialization) and RAG tenant-isolation flaws, aligned with the OWASP Top 10 for LLM Applications.
- **Web Exploitation:** Exploited SQL injection, WAF bypass, SSRF, and IDOR chaining with magic-bytes file upload bypass across 5+ CTF challenges; authored a Python PoC for CVE-2025-55182, a Next.js/React RCE vector, on Hack The Box.
- **Cryptography:** Broke ChaCha20 nonce-reuse (Hack The Box), recovered RSA keys via Hastad's Broadcast Attack, and solved classical/affine ciphers.
- **Reverse Engineering:** Bypassed anti-disassembly (ud2/SIGILL), performed IL2CPP Unity analysis with Cheat Engine, and reversed a PRNG-seeded XOR/bit-rotation cipher.
- **Network Forensics & Privilege Escalation:** Analyzed encrypted C2 traffic (RC4/AES-256-CBC) via Wireshark/tshark; chained IDOR-exposed PCAP data into FTP/SSH access and escalated Linux privileges via misconfigured capabilities (Hack The Box "Cap").
- **OSINT & Reconnaissance:** Conducted multi-platform recon with Sherlock and exiftool, JS bundle extraction on a React/Vite SPA, and git commit forensics to recover a hidden flag.

EXPERIENCE

Full Stack Developer Intern - TRANSMED Project (Remote)

Apr 2025 - Sep 2025

- Maintained a production web platform with React.js and MySQL over a 6-month engagement; delivered 10+ responsive components and integrated REST APIs with optimized MySQL queries while following secure coding practices.

PROJECTS

Image Encryption Tool

Jan 2025 - May 2025

JavaScript - Applied Cryptography

- Engineered a command-line tool that encrypts and decrypts image files using cryptographic algorithms, with robust error handling and input validation to protect sensitive visual data from unauthorized access.

Decentralized Cloud Storage System and Online Voting System

2024 - 2025

Python, JavaScript, Solidity, Web3.js, Ethereum - Blockchain Security

- Designed two blockchain-based applications, a decentralized file storage platform and an Ethereum voting system, using Solidity smart contracts for access control and tamper-proof record storage, with React.js/Node.js full-stack Web3 frontends.

AWARDS AND LEADERSHIP

- **Top 9% - TryHackMe:** Ranked among the top 9% of all users across security learning paths and CTF rooms.
- **15+ Capture The Flag Competitions (2025):** Solved real-world challenges in web exploitation, cryptography, reverse engineering, network forensics, AI/LLM security, and OSINT.
- **Top 15 Finalist - Smart India Hackathon (SIH), University Level (2024):** Recognized among the top 15 teams at the university-level round.
- **College Event Operations - BML Munjal University:** Manage multiple operations teams across college events while also working as a team member in other event teams, coordinating logistics, teamwork, and cross-team execution.

EDUCATION

Bachelor of Technology, Computer Science and Engineering

2023 - 2027

BML Munjal University, Haryana, India | CGPA: 7.1 / 10